HIGHER DIRECT IMAGES ARE COMPATIBLE WITH RESTRICTION TO AN OPEN SUBSPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The author gives two independent proofs and labels them 'First proof' and 'Second proof'; both are reproduced as written.

Editorial note. The counted spans carry 2 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-cohomology/retained-preamble.tex lines 1-81), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-localize-higher-direct-images is defined only in cohomology.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Description of higher direct images as the sheaf associated to a presheaf of cohomology. Same chapter, Lemma lemma-describe-higher-direct-images, invoked at proof lines 645-646. Supplies the first proof.
- Cohomology computed on an open subspace. Same chapter, Lemma lemma-cohomology-of-open, invoked at proof lines 645-646. Supplies the compatibility of the presheaf of cohomology with restriction to the open.
- The restriction of an injective resolution is an injective resolution of the restriction. Standard, used at proof lines 647-650. Supplies the second proof.

Source reference. Original lemma 01E1: sources/stacks-cohomology/pinned-original-source.tex, source lines 496–496. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01E4: sources/stacks-cohomology/pinned-original-source.tex, source lines 592–592. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of ringed spaces. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module. Let $V \subset Y$ be an open subspace. Denote $g : f^{-1}(V) \to V$ the restriction of f. Then we have*

$$R^p g_*(\mathcal{F}|_{f^{-1}(V)}) = (R^p f_* \mathcal{F})|_V$$

There is a similar statement for the derived image $Rf_*\mathcal{F}^\bullet$ where $\mathcal{F}^\bullet$ is a bounded below complex of $\mathcal{O}_X$-modules.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Proof. First proof. Apply Lemmas 01E4 and 01E1 to see the displayed equality. Second proof. Choose an injective resolution $\mathcal{F} \to \mathcal{I}^\bullet$ and use that $\mathcal{F}|_{f^{-1}(V)} \to \mathcal{I}^\bullet|_{f^{-1}(V)}$ is an injective resolution also. $\square$